

NMAO Build — Plain-Language Glossary

A running reference of the technical terms we use as we build, each explained simply and tied to this project. We add to it as new terms come up.

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The tools we work in

CLI (Command Line Interface) — Controlling a program by typing commands instead of clicking.

Example: `supabase functions deploy round-controller`.

Terminal / shell / bash — The text window where you type CLI commands. “Bash” is the most common shell (the language the terminal understands). Same idea as Finder/Explorer, but keyboard-driven.

Supabase — The service hosting our backend: the database, user login, file storage, and edge functions, all in one. Our project lives there.

Postgres (PostgreSQL) — The actual database engine inside Supabase. It stores every table (competitors, entries, pods, results...) and answers questions written in SQL.

Dashboard — The point-and-click website for Supabase, as opposed to the CLI. The SQL Editor and Edge Functions pages live there.

Talking to the database

SQL (Structured Query Language) — The language for asking a database to store or fetch data.

Example: `select count(*) from entries;` means “how many entries are there?”

Schema — The shape of the database: what tables exist and what columns each has. Ours is defined in the migrations folder.

Migration — One file of SQL that builds or changes the schema, numbered so they run in order.

Example: `20260805120000_tournament_engine.sql` creates the rounds/entries/pods tables.

Seed / seed data — Sample data loaded into an empty database so we have something to work with.

`seed_demo.sql` creates a fake tournament round.

Seed output — The text the seed script prints when it runs (e.g. the new `round_id`), so you know it worked and can grab values you need.

Row / record & column / field — A row is one entry in a table (one competitor, one pod). A column is one piece of info on that row (a competitor's name, rank).

UUID — A long, random, guaranteed-unique ID like `a3f8c1e2-9b0d-4f...`. We use them so two rounds (or competitors) can never share an ID.

The engine pieces

Edge function — A small program running on Supabase's servers that does work when called. Ours is `round-controller`, which runs the tournament pipeline.

Deploy — To publish your latest code so the live server runs it. If you change the function's code, you redeploy so the change takes effect.

Invoke / call — To run a deployed function. `supabase functions invoke round-controller ...`

Endpoint / URL — The web address a function answers at. Calling it is sending a request; what it sends back is the response.

POST — The type of web request that sends data to a server (as opposed to GET, which just asks for data). We POST each pipeline step.

Request body — The data attached to a request. Ours is JSON telling the function which round and step to run.

Pipeline / step — The tournament runs in ordered stages (steps): `divide → assign_judges → resolve → distribute`. Each is one job.

Idempotent — Safe to run more than once: running a finished step again does nothing instead of doubling up. Protects us from accidental double-clicks/retries.

Data formats & code

JSON (JavaScript Object Notation) — A simple text format for structured data, written as “key”: value pairs in curly braces. Example: `{"roundId":"a3f8...", "step":"divide"}`.

Variable — A named placeholder for a value. In the terminal, `RID=a3f8...` lets you write `$RID` instead of the long ID afterward.

Flag / option — An extra setting on a command, usually starting with two dashes. Example: `--body "..."` attaches data to the request.

TypeScript / TS — The programming language the engine is written in (a stricter version of JavaScript). Files end in `.ts`.

Key / token / auth — A secret string that proves who's allowed to do something. Supabase has two generations of keys: legacy JWTs (start `eyJ...`: `anon` and `service_role`) and new keys (`sb_publishable_...` and `sb_secret_...`). Our project has the new keys enabled, so the tournament function wants the **`sb_secret_...`** secret key — not the legacy `service_role`. Sending the wrong one returns ‘Invalid or expired session’. Secret keys bypass security, so they live only in your terminal or server settings, never in chat or the app.

Version control (Git / GitHub)

Git — A system that tracks every change to the code so you can see history and undo mistakes.

Repo (repository) — The project folder Git is tracking (NMAO-Tournament).

Commit — A saved snapshot of changes with a short note describing them.

Push — Upload your commits to GitHub so they're backed up and shareable.

GitHub — The website that hosts the repo online.